Symbol Reference: The Barn Effect

Niklaus Parcell

Symbol	Name	Role / meaning	Algorithm phase(s)
Dataset and setup			
N	Sample count	Total number of training samples.	Both phases; inference
p	Feature count	Number of columns ($p \geq 2$).	Both phases; inference
$[n]$	Integer range	$\{1, \ldots, n\}$; used to index features, samples, labels.	Notation throughout
$\mathcal{D}$	Dataset	Full collection $\{(\boldsymbol{x}^{(n)}, Y^{(n)})\}_{n=1}^{N}$.	Phase 2
$\boldsymbol{x}^{(n)}$	Feature vector	Row n of the dataset; $\boldsymbol{x}^{(n)} \in \mathbb{R}^p$.	Phase 2
$X_i^{(n)}$	Feature value	Value of feature i at sample n.	Phases 1 & 2
$Y^{(n)}$	Outcome	Scalar target for sample n; may be a raw label or any model's scalar output.	Phase 2; inference
$\bar{Y}$	Outcome mean	Global mean $N^{-1} \sum_n Y^{(n)}$; serves as the attribution baseline.	Inference
Chunking			
K	Chunk count	Number of contiguous data chunks; $\sum_k n_k = N$.	Phase 2
$\mathcal{D}_k$	Chunk k	The k-th subset of $\mathcal{D}$.	Phase 2
n_k	Chunk size	$\lvert\mathcal{D}_k\rvert$.	Phase 2
Discretization — Phase 1			
B_i	Bin depth	Number of interior boundaries for column i; controls bin schema granularity.	Phase 1
$b_k^{(i)}$	Bin boundary	k-th sorted boundary value for numerical column i; $b_0^{(i)} = \min$, $b_{B_i+1}^{(i)} = \max$.	Phase 1
$\mathcal{F}_i$	Frequent categories	Top-B_i most frequent distinct values of categorical column i.	Phase 1
ε	Rare-value token	Distinguished label assigned to categories not in $\mathcal{F}_i$.	Phase 1
$\mathcal{L}_i$	Label set	Finite set of bin labels produced by the bin schema for column i.	Phase 1
$\ell^{(i)}$	Label function	Map $\ell^{(i)} : \mathrm{dom}(X_i) \to \mathcal{L}_i$ assigning each value its bin label.	Phase 1; inference
$\mathcal{B}$	Bin schema	Collection $\{(\mathcal{L}_i, \ell^{(i)})\}_{i=1}^{p}$; the full discretization specification.	Phase 1
$\mathcal{V}$	Vocabulary	$\bigcup_{i=1}^{p} \mathcal{L}_i$; the set of all bin labels across all columns.	Phases 1 & 2; inference
V	Vocabulary size	$\lvert\mathcal{V}\rvert$; dimension of the square accumulator and interaction graph matrices.	Phases 1 & 2; inference
Pairwise accumulation — Phase 2			
$\mathcal{C}$	Column pair set	$\{(i,j) : 1 \leq i < j \leq p\}$; $\lvert\mathcal{C}\rvert = p(p-1)/2$.	Phase 2
(u, v)	Value pair	Encoded label pair $(\ell^{(i)}(X_i^{(n)}),\, \ell^{(j)}(X_j^{(n)})) \in [V]^2$.	Phase 2

Symbol	Name	Role / meaning	Algorithm phase(s)
α	Threshold rate	Configurable significance rate in $[0, 1)$; default $\alpha = 0.01$.	Phase 2
θ_k	Significance threshold	$\lfloor \alpha \cdot n_k \rfloor$; minimum per-chunk occurrence count for a value pair to enter the accumulator.	Phase 2
$\boldsymbol{W}$	Sum accumulator	Sparse matrix $\boldsymbol{W} \in \mathbb{R}^{V \times V}$; W_{uv} accumulates the sum of $Y^{(n)}$ over co-occurrences of (u, v). Symmetric by construction.	Phase 2
$\boldsymbol{C}$	Count accumulator	Sparse matrix $\boldsymbol{C} \in \mathbb{R}^{V \times V}$; C_{uv} accumulates the number of co-occurrences of (u, v). Symmetric by construction.	Phase 2
$\boldsymbol{\Phi}$	Interaction matrix	Sparse symmetric matrix $\boldsymbol{\Phi} \in \mathbb{R}^{V \times V}$; $\Phi_{uv} = (W_{uv}/C_{uv}) \cdot \log_{10}(C_{uv})$ on the support $C_{uv} > 1$, zero otherwise. The trained model artifact.	Phase 2; inference
Φ_{uv}	Edge weight	Log-frequency-weighted conditional mean of Y given that bin u and bin v co-occurred in the same row. Entry (u, v) of $\boldsymbol{\Phi}$; weight of edge (u, v) in G_{Φ}.	Phase 2; inference
G_{Φ}	Interaction graph	Weighted undirected graph $(\mathcal{V}, E_{\Phi}, w_{\Phi})$ with vertex set $\mathcal{V} = [V]$, edge set $E_{\Phi} = \{(u, v) : \Phi_{uv} \neq 0\}$, and $w_{\Phi}(u, v) = \Phi_{uv}$. $\boldsymbol{\Phi}$ is its weighted adjacency matrix.	Phase 2; inference
Inference and attribution			
$\psi_{ij}^{(n)}$	Pairwise score	$\Phi_{\ell^{(i)}(X_i^{(n)}),\, \ell^{(j)}(X_j^{(n)})}$; single edge lookup in $\boldsymbol{\Phi}$ for column pair (i, j) at sample n.	Inference
$d_c^{(n)}$	Active pair count	Number of column partners $j \neq c$ with a nonzero pairwise score for column c at sample n; used as the normalization denominator.	Inference
$A_c^{(n)}$	Column attribution score	Mean pairwise score for column c at sample n over its $d_c^{(n)}$ nonzero partners: ${d_c^{(n)}}^{-1} \sum_{j \neq c} \psi_{cj}^{(n)}$. Zero if $d_c^{(n)} = 0$.	Inference
$\delta_c^{(n)}$	Barn attribution	$\widetilde{\Phi}_c^{(n)} - \bar{Y}$; deviation of column c's pairwise context from the global baseline for sample n.	Inference
$\boldsymbol{\Delta}$	Attribution matrix	$N \times p$ matrix with $\Delta_{nc} = \delta_c^{(n)}$; full output of inference.	Inference
$\bar{\delta}_c$	Global importance	$N^{-1} \sum_n \delta_c^{(n)}$; mean Barn attribution for feature c across all samples; used for dataset-level feature ranking.	Inference
Analysis and extensions			
π	Pair probability	$P(\ell^{(i)}(X_i) = u,\, \ell^{(j)}(X_j) = v)$; joint probability of a value pair; appears in the convergence result.	Properties
$\boldsymbol{G}$	Gap graph	$\boldsymbol{\Phi}_{\text{data}} - \boldsymbol{\Phi}_{\text{model}}$ on shared support; identifies co-occurrences where the model deviates from empirical statistics. Requires both graphs trained on outcomes with the same scale.	Discussion
$\hat{f}$	External model	Any trained model whose scalar output $\hat{f}(\boldsymbol{x}^{(n)})$ may be used as $Y^{(n)}$ in place of the raw label.	Throughout (parametric-optional)

Symbol	Name	Role / meaning	Algorithm phase(s)
K_m	Cluster count	Number of embedding clusters for modality m in the multi-modal extension; plays the role of B_m (bin depth) for embedding-encoded columns.	Multi-modal extension
$\boldsymbol{\mu}_k^{(m)}$	Cluster centroid	Centroid of cluster k for modality m; used to define $\ell^{(m)}$ via nearest-neighbor assignment.	Multi-modal extension
Operators			
$\odot$	Hadamard product	Element-wise matrix multiplication.	Phase 2
$\oslash$	Element-wise division	Element-wise matrix division on the nonzero support.	Phase 2